

4. Phosphorus(V) chloride reacts with water to form phosphoric acid and hydrogen chloride. The reactants and products are shown in the equation below but the equation is **not** balanced.



- (a) Balance the equation. [1]
- (b) Calculate the atom economy for the formation of phosphoric acid in this reaction. [2]

Atom economy = %

5. (a) Give the meaning of the term *electronegativity*. [1]

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- (b) Explain why electronegativity increases across a period in the Periodic Table. [1]

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6. Bromine is produced commercially from the bromide ions in sea water by reaction with chlorine.

- (a) Give the **ionic** equation for this reaction. [1]

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- (b) Describe a test, apart from the use of chlorine, to show that a solution contains bromide ions. Give the reagent(s) and observation(s). [1]

Reagent(s)

Observation(s)

